

Summary for 2024-11-19-MaxFlowMinCut

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1 Summary

This Linear Programming lecture focused on the Max Flow Min Cut theorem, a fundamental concept in network optimization. The theorem states that the maximum flow achievable between a source and sink node in a network is equal to the minimum capacity of any cut separating those nodes.

* **Network Flow Problem:** This problem involves finding the maximum flow from a source node to a sink node in a directed graph, subject to capacity constraints on each edge.

* **Cut:** A cut in a network is a partition of the nodes into two sets, one containing the source and the other containing the sink. The capacity of a cut is the sum of the capacities of the edges going from the source's set to the sink's set.

* **Max Flow Min Cut Theorem:** The maximum flow from the source to the sink is equal to the minimum capacity among all possible cuts separating the source and sink.

* **Ford-Fulkerson Algorithm:** This algorithm iteratively finds augmenting paths (paths from source to sink with available capacity) and increases the flow along these paths until the maximum flow is reached. The algorithm's implementation involves labeling and scanning nodes to efficiently search for augmenting paths.

* **Linear Programming Formulation:** The maximum flow problem can be formulated as a linear program, with an objective function maximizing the flow and constraints ensuring flow conservation at each node and respecting edge capacities. A fictitious arc with infinite capacity is often added to simplify the formulation.

* **Proof using Linear Programming Duality:** The Max Flow Min Cut theorem can be proven using the duality theory of linear programming. By analyzing the complementary slackness conditions of the primal and dual linear programs, one can show that the maximum flow equals the minimum cut capacity.